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Curriculum Vitae

Profile/Summary

Applied high-energy physicist with hands-on experience in calorimetry-adjacent detector work, test beam preparation/analysis, and QA/commissioning in CERN environments. Expertise in silicon sensors and scintillators. Strong physics analysis skills with contributions to the new physics searches for the CMS collaboration. Effective in cross-team coordination and supervision within large HEP collaborations.

Education

2015–2024 **Integrated Ph.D. degree in High Energy Physics**, *Cukurova University*, Adana, Turkey
Advisors: Prof. Aysel Kayis Topaksu (Cukurova University), Assoc. Prof. Sezen Sekmen (Kyungpook National University). Thesis title: "Search for new physics with razor kinematic variables using 137 fb^{-1} of proton-proton collision UL data collected at $\sqrt{s} = 13$ TeV."

2007–2011 **B.Sc. in Physics**, *Cukurova University*, Adana, Turkey
Thesis title: "Theory of Extra Dimensions and Superstrings."

Contributions to CMS

2023–2025 **Shifter Trainer (SL and TS)**
Training and organization of CMS central data taking shifts (for Shift Leaders and Technical Shifters), working closely with CMS Run Coordination and Technical Coordination

2024–Present **HGCAL (High Granularity Calorimeter) Si Sensor Tests at CERN**

- Responsible person for the HGCAL clean room activities. Organizing and leading the sensor team for the Si sensor tests and validation in the clean room located at CERN Meyrin.
- Electronic tests and validation of HGCAL's hexagonal Si sensor in the clean room.
- Responsible for air quality monitoring systems in the clean room and the sensor storage room belonging to HGCAL.
- Responsible for migrating the sensor measurement analysis framework to lxplus9.
- Hot Cell analysis for the tested Si sensors, building a new framework

2023–2024 **Tests and installation of new ZDC module at RP area of Point-5**

- Dark current measurements of PMTs for RPD detectors of ZDC,
- Preparation of the test stand with the dark box to perform the LED test.
- Leading a small team of students to tests of ZDC and HF Phase-0 PMTs to be replaced due to the ZDC bake-out incident in 2023.
- Preparation of the analysis framework for the measurements collected from PMTs for LED and pedestal runs at the HCAL's HF test stand.
- Preparation of the Detector Note for the PMT Tests (CMS DN-2025/001)

2019–Present **Inclusive search for new physics with razor variables in pp collisions at $\sqrt{s} = 13$ TeV with full Run-II data (SUS-23-014)**

- Calculating trigger efficiency
- Estimation of Z($\rightarrow \nu\nu$) + jets background using γ+jets for photon enriched control regions
- Bin optimization for razor variables to increase the sensitivity of the signal events
- Creating data cards and computing expected upper limits on the production cross sections of various SUSY simplified models

2021 **SPS H4 beamline simulation for ECAL Test Beam**

- G4beamline simulation of H4 beamline
- Developing the analysis framework
- Preparing MC ntuple for the beamline simulation
- Preparation of material chart of the H4 beamline, simulating the part of the ECAL detector tested

2019-2020 **SPS H2 beamline simulation for HGCAL Test Beam Analysis**

- G4beamline simulation of H4 beamline
- Developing the analysis framework
- Preparing MC ntuple for the beamline simulation
- Preparation of material chart of the H4 beamline.

2018-2019 **HL-LHC and HE-LHC searches for supersymmetry in hadronic final states with boosted W bosons or top quarks using razor variables.**

- Calculating trigger efficiency,
- Scale factors calculations,
- Background estimation for W, Z, Top and QCD enriched control regions,
- Closure Test.

2016-2019 **Search for new physics using razor variables at $\sqrt{s} = 13$ TeV with boosted W and top at the final state.**

- Correct MC using various Data/MC scale factors and event reweighting
- Validating background estimation with closure tests
- Computing expected upper limits on the production cross sections of various SUSY simplified models

2016-2018 **Installation, testing and commissioning of Hadronic Forward (HF) Front-End electronics and ROBOXes.**

- Perform generic tests of Charge Integrator and Encoder (QIE10) cards,
- Assemble of the DC-DC converters and Versa Link dual optical transmitters (VTX)
- Installation of the front end electronics to the CMS at Point-5.

2018 **Brightness and uniformity measurements of plastic scintillator tiles at the CERN H2 test beam.**

- Quality assurance for the Data Acquisition

Selected Publications

2023 **CMS Collaboration**, *" Search for SUSY with boosted objects using razor variables "*, In Preparation, SUS-23-014

2020 **CMS Collaboration**, *"Construction and commissioning of CMS CE prototype silicon modules"*, JINST 16 (2021) 04, T04002

2020 **CMS Collaboration**, *"The DAQ system of the 12,000 Channel CMS High Granularity Calorimeter Prototype"*, JINST 16 (2021) 04, T04001

2019 **CMS Collaboration**, *"HL-LHC searches for new physics in hadronic final states with boosted W bosons or top quarks using razor variables"*, CMS PAS FTR-18-037

2019 **CMS Collaboration**, *"Inclusive search for supersymmetry in pp collisions at $\sqrt{s} = 13$ TeV using razor variables and boosted object identification in zero and one lepton final states"*, CMS-SUS-16-017, JHEP 03 (2019) 031

2018 **S. Chatrchyan**, *et al.*, *"Brightness and uniformity measurements of plastic scintillator tiles at the CERN H2 test beam"*, 2018 JINST 13 P01002

Talks, posters, co-authored talks

November 2025 4th DRD3 week on Solid State Detectors R&D. Talk: "Investigation of Spatial Patterns of Elevated Cell Leakage Currents in HGCAL Silicon Sensors" as co-author

January 2020 8th Beam Telescope and Test Beam (BTTB) Workshop in Tbilisi. Talk: "SPS H2 Beam Line Simulation: Energy Loss and Material Impact at HGCAL Test Beam." on behalf of CMS Collaboration

4-5 October 2018 1st National Workshop on CERN-Related Activities, Ankara. Talk: "New physics searches with razor kinematic variables with boosted top quark and W boson in the final state at 13 TeV using 2016 data." on behalf of Razor-boost team of CMS Collaboration

Computing skills

Frameworks CMSSW

Languages	Python, C++, G4beamline, BASH, R, VHDL
Libraries	ROOT, NumPy, Pandas, RooStats, TMVA
Others	Git, LaTeX
Interests	Machine Learning, AI applications

Certificates

2024	HarwardX: CS50 Introduction to Computer Science
2025	HarwardX: CS50's Introduction to Artificial Intelligence with Python
2025	Coursera: Design of Digital Circuits with VHDL Programming

Languages

Native	Turkish
Fluent	English